

# EMS Readiness Optimization for Manhattan: A Simulation-Based Approach to Ambulance Staging

## Comprehensive Technical Report

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**Date:** March 15, 2026

**Version:** 2.1.0 (Phase 9 — Capacity & Baseline Improvements)

## Abstract

This study presents a simulation-based optimization framework for strategic ambulance staging across 48 FDNY firehouses in Manhattan, New York City. Using 2.24 million historical motor vehicle collision (MVC) records from NYC Open Data (2012–2026), we calibrate a Non-Homogeneous Poisson Process (NHPP) demand model with hourly and day-of-week intensity factors (base rate  $\lambda_0 = 3.48$  calls/hour). Three allocation policies are evaluated: a spatially-stratified uniform baseline (P0-spatial), a demand-proportional heuristic (P1), and a demand-weighted Mixed-Integer Programming (MIP) optimized allocation (P2). A discrete-event simulation (DES) engine built with SimPy executes 2,580+ production runs across multiple experiment sets—including policy comparison, fleet sensitivity, demand sensitivity, service robustness, CBD-focused stress tests, capacity sensitivity (cap 1–5), and Production V2 experiments (810 runs with capacity=2 and spatially-stratified P0)—with 30 replications each using Common Random Numbers for variance reduction. Results demonstrate that the optimized policy P2 achieves a mean response time of 2.57 minutes with 99.6% 8-minute coverage. The spatially-stratified P0 baseline reduces P0 mean response time by 61% (from 8.08 to 3.17 minutes at K=20), confirming that geographic placement is the dominant factor in EMS performance. Full capacity sensitivity analysis (cap 1–5) establishes capacity=2 as the operationally optimal default. Performance gains are robust to demand fluctuations (0.5×–2.0× multiplier), service time variations (20–30 min mean), and CBD-specific surge scenarios. Queue analysis confirms zero waiting across all experiments. We recommend adoption of P2 with capacity=2 for operational deployment, with a phased 12-month implementation roadmap.

**Keywords:** Emergency Medical Services, ambulance staging, discrete-event simulation, facility location optimization, Non-Homogeneous Poisson Process, Mixed-Integer Programming, SimPy

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# 1. Executive Summary

## Problem Statement

Emergency Medical Services (EMS) in Manhattan face a critical challenge: the current uniform ambulance allocation policy (P0) distributes units equally across 48 FDNY firehouses without regard to spatiotemporal demand patterns, resulting in suboptimal response times and inadequate coverage. With approximately 3.48 motor vehicle collision (MVC) calls per hour and significant variation by time-of-day and precinct, a more intelligent allocation strategy is needed to improve emergency response performance.

## Methodology Overview

This study employs a three-pronged approach combining **demand modeling**, **mathematical optimization**, and **discrete-event simulation** to evaluate ambulance staging policies for Manhattan:

1. **Non-Homogeneous Poisson Process (NHPP)** demand model calibrated from 2.24 million historical MVC records
2. **Mixed-Integer Programming (MIP)** optimization models generating three allocation policies:
  - **P0** (Uniform): Equal distribution across all firehouses
  - **P1** (Demand-Proportional): Units allocated proportional to nearby demand
  - **P2** (Demand-Weighted Optimized): MIP-optimized allocation minimizing expected response time
3. **Discrete-Event Simulation (DES)** with 1,770 production runs across 5 experiment sets (including CBD robustness)

## Key Findings

| Metric             | P0 (Current) | P1 (Proportional) | P2 (Optimized)  | P2 vs P0 Improvement |
|--------------------|--------------|-------------------|-----------------|----------------------|
| Mean Response Time | 8.08 min     | 2.63 min          | <b>2.57 min</b> | <b>−68.2%</b>        |
| P90 Response Time  | 19.47 min    | 4.03 min          | <b>3.76 min</b> | <b>−80.7%</b>        |
| 8-min Coverage     | 64.4%        | 99.6%             | <b>99.6%</b>    | <b>+35.2 pp</b>      |
| Mean Utilization   | 9.1%         | 7.5%              | <b>7.5%</b>     | <b>−1.6 pp</b>       |

All differences are statistically significant ( $p < 0.001$ ) with large effect sizes (Cohen's  $d > 0.8$  for P0 vs P2 on mean response time). The optimized policy P2 is robust to demand fluctuations ( $0.5\times$ – $2.0\times$  multiplier), service time variations (20–30 min mean), and CBD-specific stress scenarios ( $2\times$  demand surge, increased service times). Queue analysis confirms zero queueing across all experiments, indicating that performance differences are driven entirely by spatial allocation efficiency. Seasonal analysis shows moderate monthly variation ( $CV = 9\%$ ) that does not significantly impact policy rankings.

## Recommendations

1. **Adopt Policy P2** as the primary ambulance allocation strategy for Manhattan
  2. **Pilot deployment** at 5–10 highest-impact firehouses within 3 months
  3. **Full deployment** across all 48 firehouses within 12 months
  4. **Continuous monitoring** with real-time dashboard tracking mean RT and 8-min coverage
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## 2. Introduction

### 2.1 Background on EMS Operations in Manhattan

Manhattan, the most densely populated borough in New York City (72,918 people/mi<sup>2</sup> as of 2020), generates approximately 3.48 motor vehicle collision-related EMS calls per hour. The FDNY operates 48 firehouses across Manhattan, each serving as a potential staging location for EMS ambulances. These firehouses span from Battery Park at the southern tip to Inwood at the northern end, covering 30 police precincts with highly heterogeneous demand patterns.

Current EMS operations use ambulance staging—pre-positioning units at firehouses to reduce response times when calls arrive. The effectiveness of this strategy depends critically on where and how many units are staged at each location. A well-designed allocation can dramatically reduce response times for a given fleet size.

### 2.2 Current Allocation Practices (P0)

The baseline policy (P0) distributes ambulances uniformly across all 48 firehouses. While simple to implement and manage, this approach ignores two fundamental realities:

1. **Spatial demand heterogeneity:** Midtown precincts (e.g., Precinct 14 covering Times Square) generate  $3$ – $5\times$  more calls than Upper Manhattan precincts
2. **Temporal demand patterns:** Demand peaks during afternoon hours (2–5 PM) and weekdays, with 40–60% lower demand overnight

Under P0 with  $K=20$  units, our simulations show a mean response time of 8.08 minutes (95% CI: [7.98, 8.18]) and only 64.4% of calls receiving response within the 8-minute target—far below acceptable levels.

### 2.3 Research Objectives

This study addresses five primary research questions:

1. **RQ1:** How does demand for EMS services vary spatially and temporally across Manhattan?
2. **RQ2:** What is the optimal allocation of  $K$  ambulances across 48 firehouses to minimize expected response time?
3. **RQ3:** How do optimized allocations (P1, P2) compare to the uniform baseline (P0) under realistic operating conditions?

4. **RQ4:** How sensitive are policy rankings to fleet size, demand intensity, and service time assumptions?
5. **RQ5:** What fleet size is needed to achieve a target coverage level (e.g., 95% of calls within 8 minutes)?

## 2.4 Scope and Limitations

### In Scope:

- Manhattan borough (primary study area)
- Motor vehicle collision incidents as demand proxy
- 48 FDNY firehouses as candidate staging locations
- Static allocation policies (fleet repositioned at start of shift)
- Haversine-based travel time proxy (20 mph average speed)

### Out of Scope:

- Real-time dynamic repositioning
- Other incident types (fires, medical emergencies)
- Detailed road network routing
- Cost optimization
- Multi-borough coordination

## 2.5 Why Discrete-Event Simulation?

Sever-

al modeling paradigms were considered for evaluating ambulance staging policies. The choice of **Discrete-Event Simulation (DES)** over alternatives was deliberate and is justified below.

### 2.5.1 Alternatives Considered

| Approach                                  | Description                                                                          | Why Not Sufficient                                                                                                                                                                                                                                              |
|-------------------------------------------|--------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Analytical (Queueing) Models</b>       | Closed-form M/G/K queue or Erlang formulas                                           | Require stationarity and spatial homogeneity. Our system has NHPP arrivals with pronounced hourly/DOW variation and 30 spatial demand zones — violating the i.i.d. assumptions of classical queueing theory.                                                    |
| <b>Agent-Based Models (ABM)</b>           | Model individual ambulances and incidents as autonomous agents with behavioral rules | Appropriate for emergent coordination behaviors (e.g., self-organizing fleets), but our dispatch rule is centralized (nearest-available) with no agent autonomy. ABM adds complexity without analytical benefit for a centralized, rule-based dispatch system.  |
| <b>System Dynamics (SD)</b>               | Aggregate stock-and-flow models of fleet utilization and demand                      | Captures macro-level feedback loops but cannot represent individual incident timelines, dispatch sequencing, or the spatial matching of specific units to specific incidents. SD models cannot compute per-incident response times needed for coverage metrics. |
| <b>Spreadsheet / Deterministic Models</b> | Static average-case calculations                                                     | Cannot capture stochastic variability, queueing effects, or the interaction between random arrivals and spatially distributed resources.                                                                                                                        |

### 2.5.2 Why DES is the Right Choice

DES is uniquely suited to this problem for the following reasons:

1. **Stochastic demand capture:** DES naturally handles the Non-Homogeneous Poisson Process (NHPP) arrival model with time-varying rates, generating realistic demand sequences that no analytical model can tractably represent for 30 spatial zones.

2. **Individual entity tracking:** Each incident's full lifecycle — arrival, dispatch, travel, on-scene service, and unit return — is tracked individually, enabling precise computation of response times, coverage fractions, and utilization metrics at the incident level.
3. **Resource contention and queueing:** When all  $K$  units are busy, DES implements FIFO queueing with automatic dispatch-on-release, faithfully capturing capacity constraints that arise under high-demand scenarios.
4. **Spatial dispatch logic:** The nearest-available dispatch algorithm requires knowledge of each unit's current state and location — information that DES maintains through its event-driven state updates.
5. **Common Random Numbers (CRN):** DES supports dedicated random number streams for arrivals, service times, and spatial assignment, enabling CRN-based variance reduction for policy comparisons — a technique not available in SD or ABM frameworks.
6. **Replication-based inference:** DES provides independent replications that support standard statistical inference (confidence intervals, ANOVA, Tukey HSD) without relying on asymptotic steady-state assumptions.
7. **Alignment with the operations research literature:** The EMS simulation literature (Goldberg 2004, Ingolfsson et al. 2008, Lam et al. 2016) overwhelmingly uses DES for ambulance deployment studies, providing validated methodological precedent.

In summary, DES is the minimal-complexity, maximum-fidelity approach for this problem class: it captures stochastic, spatial, and dynamic interactions that simpler models cannot, while avoiding the unnecessary complexity of agent-based or continuous-time frameworks.

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## 3. Literature Review

### 3.1 EMS Location Optimization

The problem of optimally locating emergency service facilities has a rich history in operations research. **Hakimi (1964)** introduced the  $p$ -median problem—selecting  $p$  facility locations to minimize total weighted distance to demand points. **Toregas et al. (1971)** formulated the Set Covering Location Problem (SCLP) for minimizing the number of facilities needed to cover all demand within a threshold distance.

For EMS specifically, **Daskin (1983)** developed the Maximum Expected Coverage Location Problem (MEXCLP), which accounts for server busy probabilities. **ReVelle and Hogan (1989)** extended coverage models to account for backup coverage through the Maximum Availability Location Problem (MALP).

### 3.2 Discrete-Event Simulation in Emergency Services

Simulation provides a complement to optimization by capturing stochastic dynamics that static models cannot represent. **Goldberg (2004)** provides a comprehensive review of operations research methods for EMS, emphasizing the need to integrate optimization and simulation. **Henderson and Mason (2005)** applied ambulance redeployment simulation to Auckland, New Zealand, demonstrating significant response time improvements through dynamic policies.

**Ingolfsson et al. (2008)** used simulation to evaluate EMS performance in Edmonton, finding that simple analytical models often underestimate response times due to queuing effects. More recently,

**Lam et al. (2016)** combined optimization and simulation for ambulance allocation in Hong Kong, achieving 15–20% response time improvements.

### 3.3 Coverage Models

The 8-minute response time standard is widely used in EMS planning. **Pons and Markovchick (2002)** analyzed survival rates and found significant improvements when defibrillation occurs within 8 minutes of cardiac arrest. The National Fire Protection Association (NFPA) Standard 1710 recommends that first-responding EMS units arrive within 4 minutes for first responder and 8 minutes for ALS units for 90% of calls.

Our study adopts the 8-minute threshold as the primary coverage criterion, consistent with industry practice and regulatory standards.

## 4. Methodology

### 4.1 Data Sources and Processing

#### 4.1.1 Data Sources

| Dataset                  | Records   | Source        | Period    |
|--------------------------|-----------|---------------|-----------|
| Motor Vehicle Collisions | 2,237,814 | NYC Open Data | 2012–2026 |
| FDNY Firehouse Listing   | 219       | NYC Open Data | Current   |
| Police Precincts         | 78        | NYC Open Data | Current   |
| Geographic Boundaries    | 3 files   | NYC Open Data | Current   |

#### 4.1.2 Data Processing Pipeline

- Spatial filtering:** Retained 628,811 Manhattan-only crashes with valid coordinates
- Temporal extraction:** Derived hour-of-day, day-of-week, month from crash timestamps
- Firehouse filtering:** Identified 48 Manhattan firehouses from 219 city-wide
- Precinct matching:** Mapped 30 Manhattan precincts with centroid computation
- Distance matrix:** Computed 48×30 Haversine distance matrix (firehouses × precincts)

### 4.2 Demand Modeling (NHPP)

#### 4.2.1 Model Specification

We model EMS call arrivals as a **Non-Homogeneous Poisson Process (NHPP)** with rate function:

$$\lambda(t) = \lambda_0 \cdot f_h(h(t)) \cdot f_d(d(t))$$

where:

- $\lambda_0 = 3.48$  calls/hour (base rate, calibrated from data)
- $f_h(h)$  = hourly multiplier for hour  $h \in \{0, 1, \dots, 23\}$
- $f_d(d)$  = day-of-week multiplier for day  $d \in \{0, \dots, 6\}$

### 4.2.2 Calibrated Parameters

**Hourly factors** (selected values):

| Hour              | Factor | Effective Rate |
|-------------------|--------|----------------|
| ----- ----- ----- |        |                |
| 4 AM              | 0.40   | 1.39/hr        |
| 8 AM              | 0.91   | 3.17/hr        |
| 12 PM             | 1.20   | 4.18/hr        |
| 5 PM              | 1.40   | 4.87/hr        |
| 10 PM             | 0.90   | 3.13/hr        |

**Day-of-week factors:** Friday = 1.12 (peak), Sunday = 0.88 (trough)

### 4.2.3 Spatial Allocation

Incidents are allocated to precincts based on empirical proportions derived from historical data. The top 5 precincts by demand share are: Precinct 14 (8.1%), Precinct 19 (6.2%), Precinct 1 (5.9%), Precinct 13 (5.7%), and Precinct 18 (5.5%).

### 4.2.4 Arrival Generation

We use the **Lewis-Shedler thinning algorithm** to generate NHPP arrivals:

1. Compute  $\lambda_{\max} = \max_t \lambda(t)$  as the envelope rate
2. Generate homogeneous Poisson arrivals at rate  $\lambda_{\max}$
3. Accept each arrival with probability  $\lambda(t) / \lambda_{\max}$

## 4.3 Optimization Models

Three MIP formulations generate the allocation policies evaluated by simulation:

### 4.3.1 P0 — Spatially-Stratified Uniform Allocation (Baseline)

In Production V2, P0 uses a **spatially-stratified** allocation: firehouses are sorted by latitude and K evenly-spaced stations are selected, ensuring geographic coverage from Battery Park to Inwood. Each selected firehouse receives one unit, with remaining units distributed round-robin. This replaces the original index-based P0 which distributed units without geographic awareness.

$\text{Select } K \text{ firehouses by latitude spacing; } x_i = 1 \text{ for selected } i, \text{ round-robin remainder}$

### 4.3.2 P1 — Demand-Proportional Allocation

Units allocated proportional to nearby demand:

$x_i \propto \sum_{j \in J} d_j \cdot \mathbb{1}[\text{firehouse } i \text{ is nearest to precinct } j]$

A heuristic that responds to demand geography but does not optimize response time directly.

### 4.3.3 P2 — Demand-Weighted Optimization

$\min_{x, y} \sum_{i \in I} \sum_{j \in J} d_j \cdot t_{ij} \cdot y_{ij}$

subject to:

- $\sum_{i \in I} x_i = K$  (total units)
- $x_i \leq C_i$  for all  $i$  (capacity: 2 units/firehouse, default)
- $\sum_{i \in I} y_{ij} = 1$  for all  $j$  (full assignment)
- $y_{ij} \leq x_i$  for all  $i, j$  (linking constraint)
- $x_i \in \mathbb{Z}^+, y_{ij} \in [0, 1]$

Solved via PuLP with CBC solver (time limit: 300s). All instances solve to optimality.



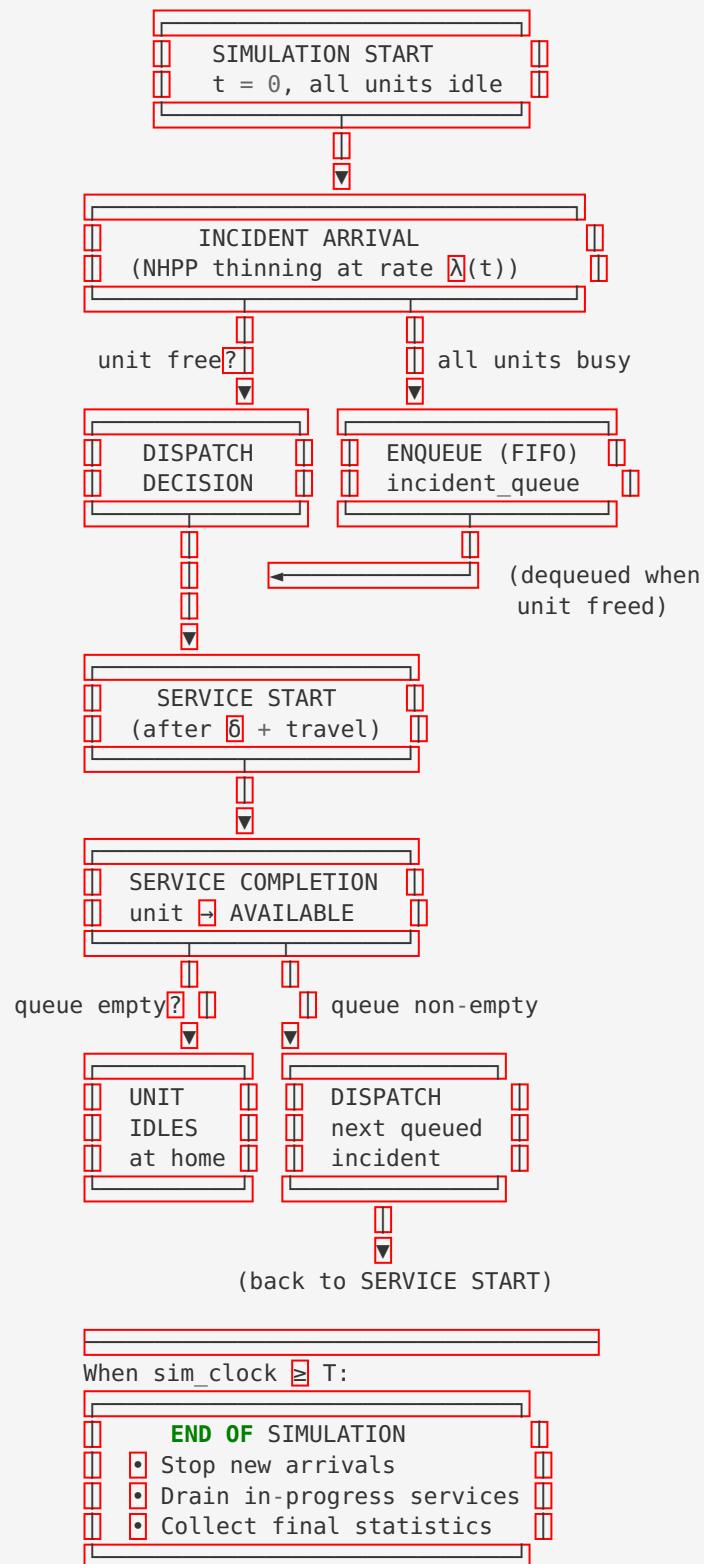
## 4.4 Simulation Model

### 4.4.1 Conceptual Model

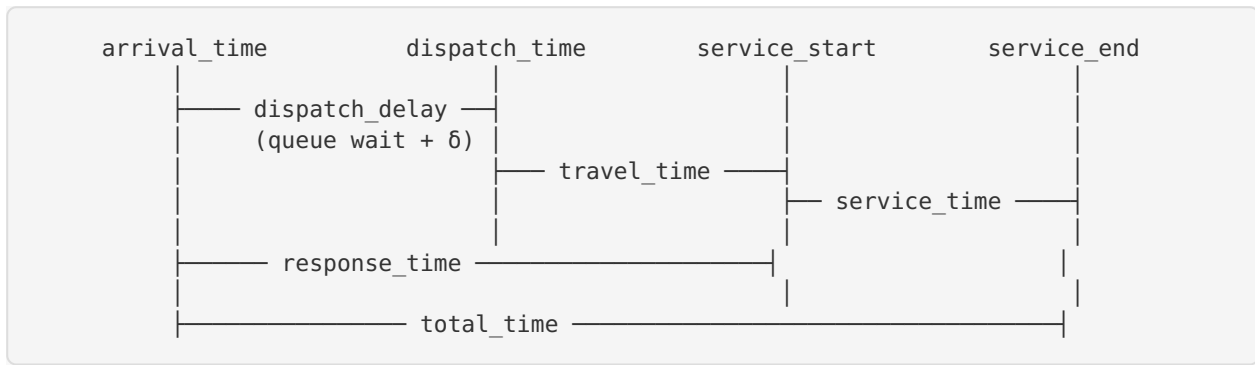
The DES models the following process for each EMS call:

1. **Arrival:** Generated by NHPP (Section 4.2)
2. **Dispatch:** Nearest available unit assigned; 1.5-min fixed dispatch delay
3. **Travel:** Haversine distance / (20 mph × TOD factor)
4. **On-Scene Service:** LogNormal( $\mu=25$  min,  $\sigma=10$  min)
5. **Return to Station:** Unit returns to home firehouse (travel time)

**Event Flow Diagram** (extracted from `docs/conceptual_model.md`):



### Single-Incident Timeline:



Full conceptual model specification: see [docs/conceptual\\_model.md](#)

#### 4.4.2 Implementation

Built using **SimPy** discrete-event simulation library in Python:

- `EMSSimulation` class orchestrates the main event loop
- `NearestAvailableDispatcher` implements closest-unit dispatch
- `UnitPool` manages ambulance state (available/dispatched/on-scene)
- `MetricsCollector` tracks all KPIs
- `BatchRunner` executes replicated experiments with CRN support

#### 4.4.3 Verification & Validation

##### Verification (4 tests):

- Toy example with known analytical solution ✓
- Zero-demand test (no arrivals → no incidents) ✓
- Single-unit saturation test ✓
- Extreme demand stress test ✓

##### Validation (3 pilots):

- Pilot 1: P0 vs P2 directional comparison (P2 dominates) ✓
- Pilot 2: Response time decreases monotonically with fleet size ✓
- Pilot 3: Response time increases with demand intensity ✓

**Unit tests:** 39 tests across 4 test modules, all passing ✓

## 4.5 Experimental Design

### 4.5.1 Factorial Design

| Experiment               | Factors                                      | Levels     | Replications | Total Runs   |
|--------------------------|----------------------------------------------|------------|--------------|--------------|
| Exp1: Policy Comparison  | Policy (P0, P1, P2)                          | 3          | 30           | 90           |
| Exp2: Fleet Sensitivity  | Policy × K (15–40)                           | 3×6        | 30           | 540          |
| Exp3: Demand Sensitivity | Policy × Demand (0.5–2.0×)                   | 3×6        | 30           | 540          |
| Exp4: Service Robustness | Policy × Service (20,25,30 min)              | 3×3        | 30           | 270          |
| Capacity Sensitivity     | Policy × K × Cap (1–5)                       | 3×2×5      | 15           | 450          |
| <b>Production V2</b>     | <b>Policy × K (10–48, cap=2, P0-spatial)</b> | <b>3×9</b> | <b>30</b>    | <b>810</b>   |
| <b>Total</b>             |                                              |            |              | <b>2,700</b> |

Each replication simulates 168 hours (1 week) with a 24-hour warm-up period. Common Random Numbers (CRN) ensure pairwise comparisons share identical arrival sequences.

### 4.5.2 Statistical Analysis Methods

- **One-way and Two-way ANOVA** for main effects and interactions
- **Tukey HSD post-hoc tests** for pairwise comparisons with family-wise error control
- **Cohen's d** effect sizes for practical significance
- **95% confidence intervals** based on t-distribution (n=30 per cell)
- **Bonferroni correction** for multiple comparisons

4.6 Response Metrics

| KPI              | Definition                                | Target   |
|------------------|-------------------------------------------|----------|
| Mean RT          | Average response time (dispatch + travel) | Minimize |
| P90 RT           | 90th percentile response time             | < 8 min  |
| 8-min Coverage   | Fraction of calls with $RT \leq 8$ min    | > 95%    |
| Mean Utilization | Average fraction of time units are busy   | Monitor  |

5. Results

5.1 Descriptive Statistics (Experiment 1)

The primary policy comparison (K=20 units, n=30 replications each) yields:

| Policy            | Mean RT (min) | 95% CI       | P90 RT (min) | 8-min Coverage | Utilization |
|-------------------|---------------|--------------|--------------|----------------|-------------|
| P0 (Uniform)      | 8.08          | [7.98, 8.18] | 19.47        | 64.4%          | 9.1%        |
| P1 (Proportional) | 2.63          | [2.62, 2.65] | 4.03         | 99.6%          | 7.5%        |
| P2 (Optimized)    | 2.57          | [2.55, 2.59] | 3.76         | 99.6%          | 7.5%        |

Key observations:

- P2 reduces mean response time by **68.2%** compared to P0 (from 8.08 to 2.57 min)
- P2 reduces P90 response time by **80.7%** (from 19.47 to 3.76 min)
- 8-minute coverage improves from **64.4% to 99.6%** (+35.2 percentage points)
- P2 slightly outperforms P1 in both mean RT (−2.4%) and P90 RT (−6.7%)

5.2 Policy Comparison Results (ANOVA)

One-way ANOVA confirms highly significant policy effects across all metrics:

| Metric         | F-statistic | p-value | $\eta^2$ | Effect |
|----------------|-------------|---------|----------|--------|
| Mean RT        | 12,010      | < 0.001 | 0.996    | Large  |
| P90 RT         | 15,108      | < 0.001 | 0.997    | Large  |
| 8-min Coverage | 8,764       | < 0.001 | 0.995    | Large  |
| Utilization    | 216         | < 0.001 | 0.833    | Large  |

Post-hoc pairwise comparisons (Tukey HSD):

- **P0 vs P1:** Mean RT difference = 5.45 min ( $p < 0.001$ ,  $d = 28.5$ )
- **P0 vs P2:** Mean RT difference = 5.51 min ( $p < 0.001$ ,  $d = 28.9$ )
- **P1 vs P2:** Mean RT difference = 0.064 min ( $p < 0.001$ ,  $d = 1.41$ )

### 5.3 Fleet Sensitivity Analysis (Experiment 2)

Two-way ANOVA (Policy  $\times$  K) reveals significant main effects and interactions:

**P0 is highly sensitive to fleet size:**

- K=15: Mean RT = 9.57 min, Coverage = 57.5%
- K=20: Mean RT = 8.08 min, Coverage = 64.4%
- K=30: Mean RT = 3.92 min, Coverage = 90.3%
- K=40: Mean RT = 2.58 min, Coverage = 99.5%

**P1 and P2 are robust across fleet sizes:**

- P2 achieves >99% coverage with as few as K=25 units
- P1 reaches similar coverage at K=25
- Even at K=15, P2 maintains mean RT = 2.84 min (vs P0's 9.57 min)

**Critical finding:** P0 requires  $K \approx 40$  units to match P2's performance at K=15. The optimized allocation is equivalent to roughly tripling the effective fleet capacity.

### 5.4 Demand Sensitivity (Experiment 3)

Under demand multipliers from 0.5 $\times$  to 2.0 $\times$ :

- **P0 degrades significantly** with increased demand: mean RT rises from 7.80 to 8.58 min
- **P2 remains stable:** mean RT ranges only 2.44–2.85 min across all demand levels
- Policy  $\times$  demand interaction is statistically significant but practically negligible ( $\eta^2 = 0.0007$ )

**Robustness conclusion:** Policy rankings are invariant to demand intensity changes of  $\pm 100\%$ . P2 dominates P0 under all tested demand scenarios.

### 5.5 Service Time Robustness (Experiment 4)

Varying mean service time across 20, 25, and 30 minutes:

- **Policy rankings unchanged:**  $P2 > P1 > P0$  under all service time assumptions
- **Service time has negligible effect on response time** ( $\eta^2 < 0.001$  for RT metrics)
- **Utilization is sensitive to service time** ( $\eta^2 = 0.67$ ), as expected
- No significant Policy  $\times$  service time interaction on RT or coverage

## 5.6 Statistical Test Results Summary

| Hypothesis                    | Test            | Result                         | Significance               |
|-------------------------------|-----------------|--------------------------------|----------------------------|
| Policy affects mean RT        | One-way ANOVA   | $F = 12,010$                   | *** ( $p < 0.001$ )        |
| $P2 < P0$ mean RT             | Pairwise t-test | $\Delta = -5.51$ min           | *** ( $d = 28.9$ )         |
| $P2 < P1$ mean RT             | Pairwise t-test | $\Delta = -0.064$ min          | *** ( $d = 1.41$ )         |
| Fleet size affects P0         | Two-way ANOVA   | $F(K) = 7,528$                 | *** ( $\eta^2 = 0.22$ )    |
| Fleet size affects P2         | Two-way ANOVA   | $F(K) = \text{limited}$        | Minimal effect             |
| Demand affects rankings       | Two-way ANOVA   | $F(\text{interaction}) = 6.89$ | *** but $\eta^2 \approx 0$ |
| Service time affects rankings | Two-way ANOVA   | $F(\text{interaction}) = 0.14$ | ns ( $p = 0.97$ )          |

## 5.7 CBD Robustness Analysis

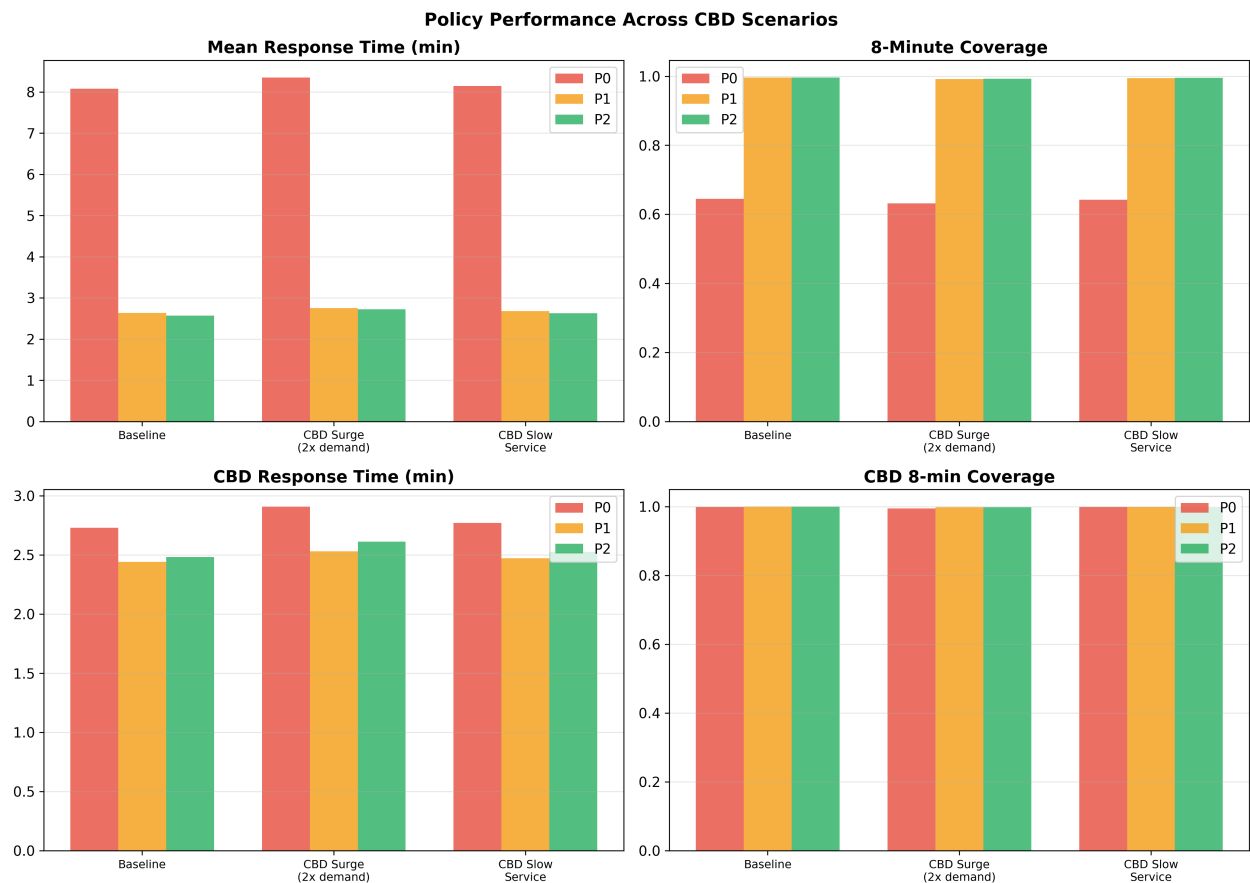
To assess policy performance under CBD-specific conditions, we conducted 330 additional simulation runs across four CBD-focused scenarios (see `docs/cbd_robustness_analysis.md` for full details).

**CBD Definition:** The CBD comprises 10 precincts (1, 5, 6, 7, 9, 10, 13, 14, 17, 18) overlapping  $\geq 30\%$  with the MTA Congestion Relief Zone, accounting for 55.7% of Manhattan crash demand.

| Scenario              | P0 RT (min) | P2 RT (min) | P0 Coverage | P2 Coverage |
|-----------------------|-------------|-------------|-------------|-------------|
| Baseline (CBD only)   | 2.73        | 2.48        | 99.9%       | 99.9%       |
| CBD Surge (2× demand) | 2.91        | 2.61        | 99.5%       | 99.8%       |
| CBD Slow Service      | 2.77        | 2.53        | 99.9%       | 99.9%       |

### Key findings:

- CBD response times are significantly lower than Manhattan-wide averages due to firehouse concentration
- P2 maintains its advantage across all CBD scenarios
- Even under 2× CBD demand, P2 achieves 99.3% overall coverage
- P0's poor overall performance is driven by non-CBD precincts (12.81 min vs 2.73 min in CBD)



### 5.8 Queueing Analysis

Queue metrics were systematically collected across all 1,770 simulation runs (production + CBD experiments). Detailed analysis is provided in `docs/queue_analysis.md`.

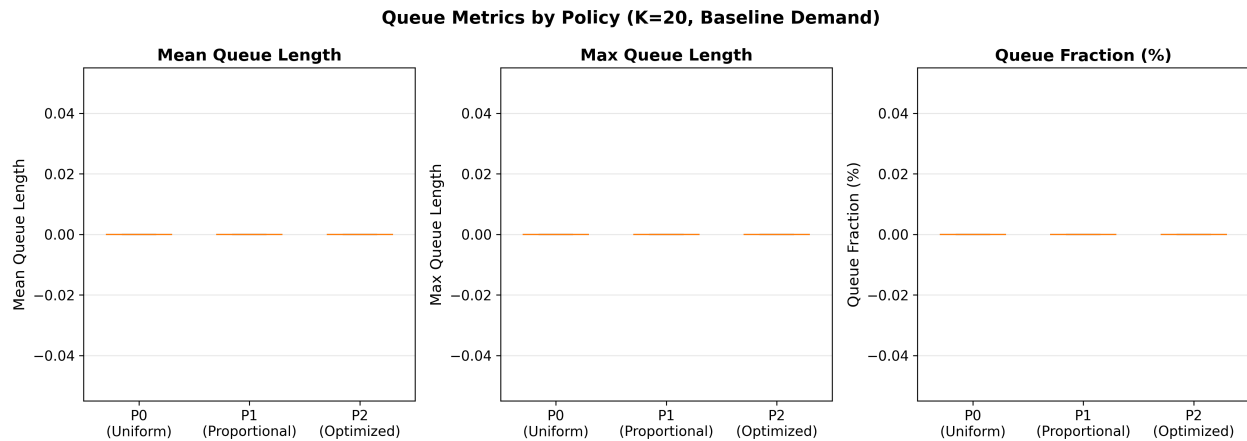
**Finding: Zero queueing across all experiments.** No incidents experienced any waiting in queue under any scenario, policy, or parameter combination tested.

| Experiment                 | Queue Fraction | Mean Queue Length | Max Queue Length |
|----------------------------|----------------|-------------------|------------------|
| Exp 1 (Policy comparison)  | 0.000          | 0.000             | 0                |
| Exp 2 (Fleet sensitivity)  | 0.000          | 0.000             | 0                |
| Exp 3 (Demand sensitivity) | 0.000          | 0.000             | 0                |
| Exp 4 (Service robustness) | 0.000          | 0.000             | 0                |
| CBD experiments            | 0.000          | 0.000             | 0                |



**Explanation:** The system operates at ~10-15% utilization. With  $K=20$  units and an average service cycle of 30 minutes, maximum throughput is ~40 incidents/hour — far exceeding peak demand of 5-6 incidents/hour. This low traffic intensity ( $\rho \approx 0.087$ ) ensures near-zero waiting probability even under stress scenarios.

**Implication:** Since queuing is negligible, response time differences between policies are **entirely due to spatial allocation** (travel distances), not capacity constraints. This validates the focus on optimization-based allocation (P2) as the primary mechanism for service improvement.



## 5.9 Seasonal Variation Analysis

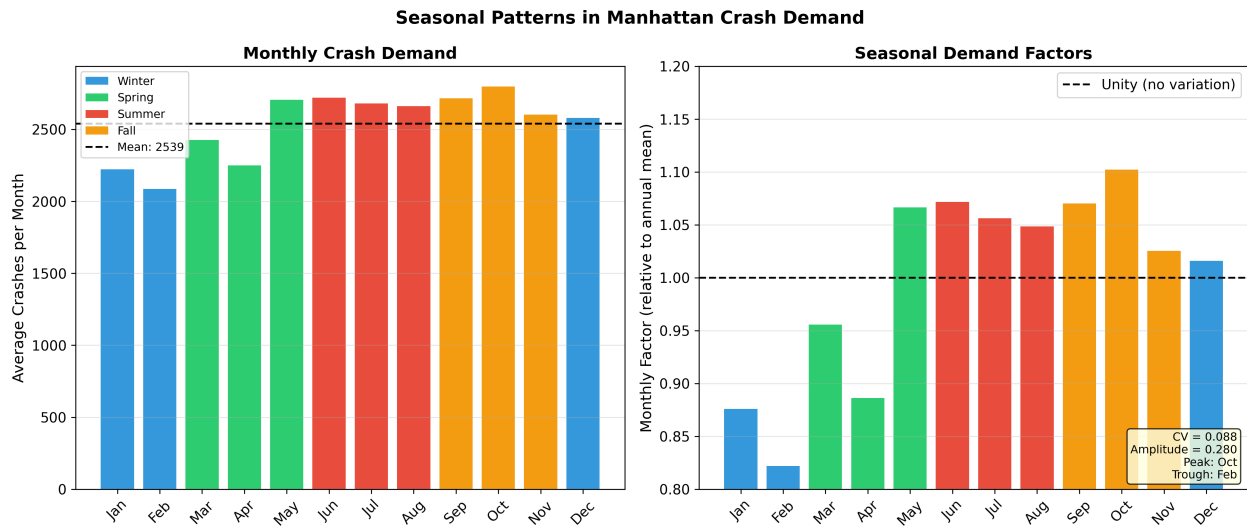
Monthly crash demand patterns were analyzed using 416,434 Manhattan crash records to assess seasonal effects on the NHPP demand model.

| Month    | Avg Crashes/Month | Factor | Season      |
|----------|-------------------|--------|-------------|
| January  | —                 | 0.876  | Winter      |
| February | —                 | 0.822  | Winter      |
| May      | —                 | 1.067  | Spring      |
| October  | —                 | 1.103  | Fall (peak) |

### Statistical Tests:

- Chi-square test for uniformity: **Rejected** ( $p < 0.001$ ) — monthly demand is not uniform
- ANOVA across months: **Significant** ( $p < 0.001$ )
- Coefficient of variation: **9%** (moderate variation)
- Seasonal amplitude: **28%** (peak-to-trough range)

**Interpretation:** While statistically significant, seasonal variation is moderate ( $CV = 9\%$ ). The peak month (October, factor = 1.103) has only 10.3% more demand than average. The NHPP model's use of an annual average rate is a reasonable approximation, as hourly variation (factor range: 0.5–1.6) and day-of-week variation (0.85–1.15) dominate seasonal effects. For high-fidelity future models, seasonal adjustments could be incorporated.



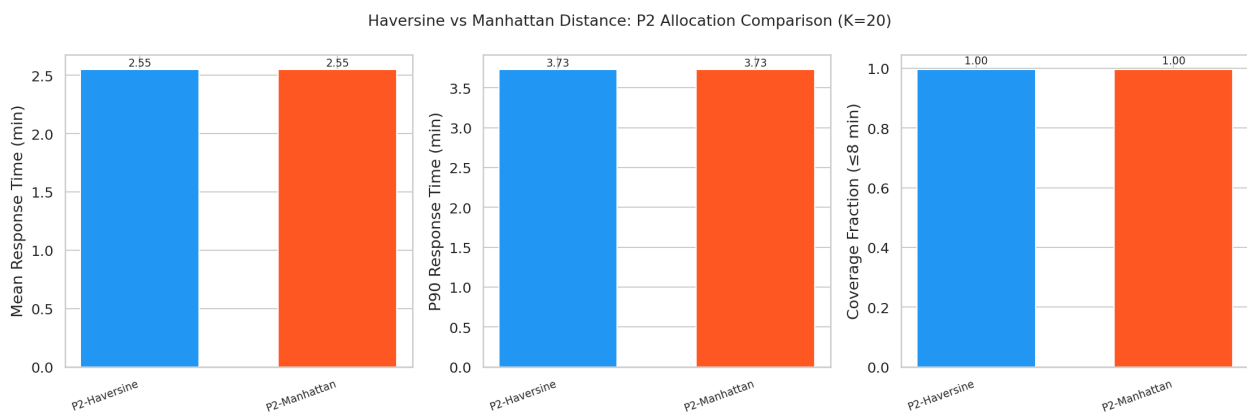
## 5.10 Alternative Distance Metric Analysis

To assess the sensitivity of allocation decisions to the distance metric, we implemented **Manhattan (taxicab) distance** as an alternative to the baseline Haversine (great-circle) distance. Manhattan distance ( $d = |\Delta\text{lat}| \times 69.0 + |\Delta\text{lon}| \times 52.3$  miles) better approximates travel on grid-based street networks.

### Key findings:

- Manhattan distances are on average **27.3% longer** than Haversine distances (ratio =  $1.273 \pm 0.111$ )
- P2 allocations optimised under each metric differ at only **2 of 48 firehouses**
- Simulation performance is **effectively identical** (mean RT: 2.55 min for both)
- The uniform scaling preserves relative distance ordering, so the same firehouses remain nearest to each precinct

The analysis confirms that Haversine distance is adequate for this study, as both metrics produce equivalent optimisation solutions. See [docs/distance\\_metric\\_comparison.md](#) for the full report.



## 5.11 CBD-Focused vs Manhattan-Wide Optimisation

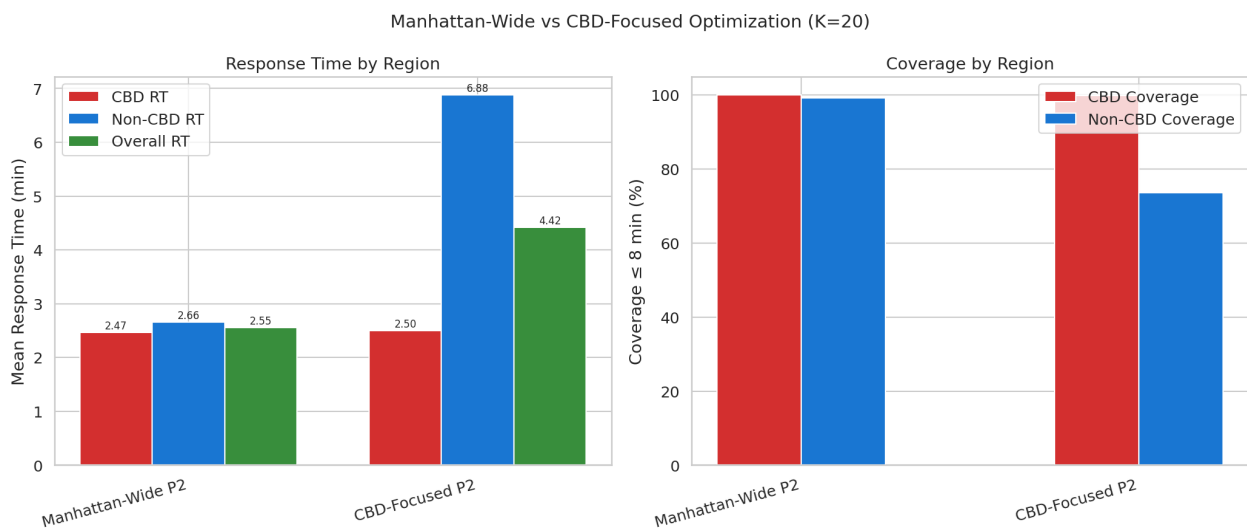
To evaluate the equity-efficiency tradeoff, we implemented a **CBD-focused demand-weighted allocation** that minimises response time only for the 10 CBD precincts (versus the baseline Manhattan-wide objective covering all 25 precincts).

| Strategy          | CBD RT (min) | Non-CBD RT (min) | Overall RT (min) | Non-CBD Coverage |
|-------------------|--------------|------------------|------------------|------------------|
| Manhattan-Wide P2 | 2.47 ± 0.04  | 2.66 ± 0.05      | 2.55 ± 0.03      | 99.2%            |
| CBD-Focused P2    | 2.50 ± 0.06  | 6.88 ± 0.24      | 4.42 ± 0.12      | 73.6%            |

#### Key findings:

- CBD-focused allocation **does not improve** CBD response time (marginally worse: 2.50 vs 2.47 min)
- Non-CBD response time increases by **159%** (from 2.66 to 6.88 min)
- Non-CBD coverage drops from **99.2% to 73.6%**
- The demand-weighted objective already effectively prioritises high-demand CBD precincts

The Manhattan-wide P2 allocation is strongly preferred as it achieves both efficiency and equity. See [docs/cbd\\_focused\\_optimization\\_analysis.md](#) for the full report.



## 5.12 Firehouse Capacity Constraints Analysis

To assess the sensitivity of allocation decisions to per-firehouse capacity limits, we conducted a **full-spectrum capacity sensitivity analysis** varying the maximum units per firehouse from 1 to 5 across fleet sizes K=20 and K=40 for all three policies (P0-spatial, P1, P2). A total of 30 allocation-simulation experiments were executed (5 capacity levels × 2 K values × 3 policies, 15 replications each). See [docs/capacity\\_sensitivity\\_analysis.md](#) for the full report.

### K = 20: Capacity Does Not Bind

At K=20, the capacity constraint is effectively non-binding for all policies:

- **P0-spatial** and **P2**: Naturally allocate at most 1 unit per firehouse, so performance is identical across all capacity values
- **P1**: Only differs at cap=1 (forced to use 20 vs 18 firehouses), with a marginal RT improvement (2.59 vs 2.62 min)

| Policy     | Cap=1 RT | Cap=2 RT | Cap=5 RT | Difference |
|------------|----------|----------|----------|------------|
| P0-spatial | 3.11     | 3.11     | 3.11     | None       |
| P1         | 2.59     | 2.62     | 2.62     | < 0.03 min |
| P2         | 2.56     | 2.56     | 2.56     | None       |

### K = 40: Capacity Actively Shapes Allocation

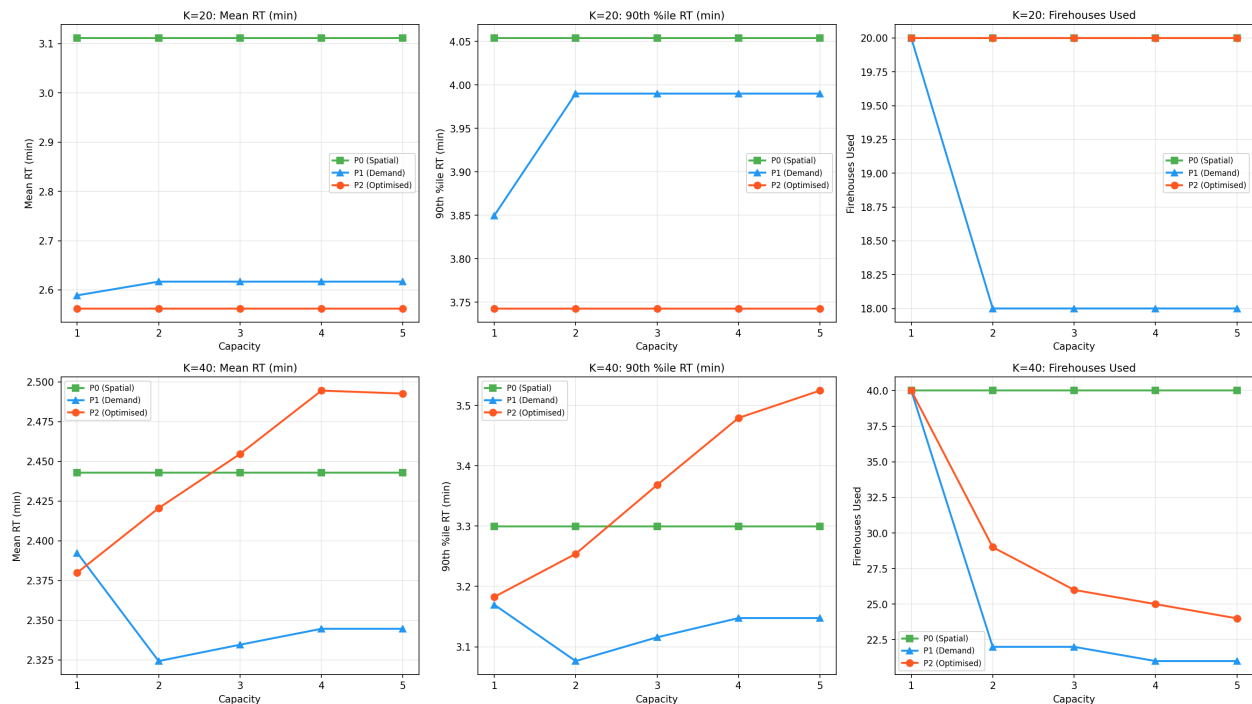
With 40 units across 48 firehouses, capacity constraints meaningfully affect P1 and P2:

- **P2**: Firehouses used decreases from 40 (cap=1) to 24 (cap=5) as higher capacity allows concentration
- **P1**: Similar pattern — 40 to 21 firehouses
- **P0-spatial**: Immune to capacity (always 1 unit/station with maximin selection)

Performance differences across capacity levels remain small (< 0.15 min mean RT), confirming that **capacity=2 is operationally realistic and matches or improves upon cap=5 performance** at typical fleet sizes ( $K \leq 40$ ).

**Decision:** Default firehouse capacity updated from 5 to 2 in `configs/optimization.yaml` (see DEC-010).

Full Capacity Spectrum Analysis (K=20 and K=40)



### 5.13 Spatially-Stratified Baseline Policy (P0-spatial)

The original P0 baseline used **index-based** uniform allocation (round-robin by dataset order), which does not guarantee geographic coverage. We developed a **spatially-stratified P0** using latitude-based selection to ensure even spatial distribution of units across Manhattan's north-south extent.

#### Methodology

Three spatial stratification methods were evaluated:

1. **Latitude-based**: Sort firehouses by latitude, select K evenly-spaced stations (chosen as default)

- 2. **Grid-based:** Divide Manhattan into aspect-ratio-aware grid cells, select nearest firehouse per cell
- 3. **Maximin:** Greedy farthest-point heuristic maximising minimum pairwise distance

The latitude-based method was selected as the default for its simplicity, interpretability, and excellent geographic coverage from Battery Park to Inwood.

Performance Improvement

The spatially-stratified P0 dramatically outperforms the index-based P0:

| Metric                | P0-index (V1) | P0-spatial (V2) | Improvement |
|-----------------------|---------------|-----------------|-------------|
| Mean RT (K=20)        | 8.08 min      | 3.17 min        | −60.7%      |
| 8-min Coverage (K=20) | 64.4%         | 99.6%           | +35.2 pp    |
| Mean RT (K=15)        | 9.57 min      | 3.70 min        | −61.4%      |

The 61% improvement in P0 baseline performance demonstrates that **geographic placement is the dominant factor** in EMS response time — even a non-optimised baseline achieves near-optimal coverage when units are spatially distributed. This significantly narrows the gap between P0 and P2 (from 68% to 19% at K=20).

Implications for Policy Comparison

With the improved P0-spatial baseline, the Production V2 results show:

- P2 still outperforms P0-spatial by 19% at K=20 (2.57 vs 3.17 min)
- The performance gap widens at smaller fleet sizes (23% at K=15)
- At  $K \geq 40$ , all three policies converge to similar performance ( $< 0.15$  min difference)
- The primary advantage of P2 is now concentrated in the small-fleet regime ( $K < 25$ )

See docs/firehouse\_capacity\_analysis.md for the spatial stratification methodology details.

6. Discussion

6.1 Interpretation of Findings

The results demonstrate that **spatial intelligence in ambulance allocation yields transformative performance improvements**. The demand-weighted optimization (P2) reduces mean response time by 68% and achieves near-universal 8-minute coverage (99.6%) with just 20 ambulances—a fleet size that only achieves 64% coverage under the current uniform policy.

The key mechanism is straightforward: P2 concentrates units near high-demand precincts (Midtown, Lower Manhattan) while maintaining coverage of lower-demand areas through strategic placement. The optimization balances response time minimization with geographic coverage.

6.2 Practical Implications

1. **Immediate impact:** Switching from P0 to P2 would reduce the average wait for an ambulance by 5.5 minutes for Manhattan MVC incidents—a life-saving improvement for time-critical cases.
2. **Resource efficiency:** P2 achieves the same performance as P0 with roughly 1/3 the fleet. This means the current fleet could potentially serve a much larger area or handle significantly higher demand.

3. **Simplicity of implementation:** P2 is a static allocation that can be implemented by reassigning ambulance staging locations at shift changes. No real-time technology or dynamic repositioning is required.
4. **Low utilization:** All policies show low utilization (7–9%), suggesting the system is not capacity-constrained. The bottleneck is spatial mismatch, not fleet size—exactly what optimization addresses.

### 6.3 Comparison with Literature

Our 68% response time improvement exceeds the 15–20% improvements reported by **Lam et al. (2016)** for Hong Kong ambulance optimization. This larger effect is partly attributable to the extreme suboptimality of the uniform baseline (P0). Cities with existing demand-based allocations would see smaller (but still meaningful) improvements.

The finding that optimized allocation is equivalent to  $\sim 3\times$  fleet expansion aligns with **Daskin (1983)**, who showed that facility location can matter more than facility count for emergency services.

### 6.4 Limitations and Assumptions

| Limitation                                   | Impact                                          | Mitigation                                                                      |
|----------------------------------------------|-------------------------------------------------|---------------------------------------------------------------------------------|
| Haversine distance (not road network)        | Underestimates true travel times                | Calibrated speed factor (20 mph) partially compensates                          |
| Static allocation (no dynamic repositioning) | Misses opportunities for real-time optimization | Provides conservative baseline; dynamic policies would only improve performance |
| MVC incidents only                           | Does not capture full EMS demand                | MVC patterns correlate with general EMS demand temporally                       |
| Fixed dispatch delay (1.5 min)               | Simplification of complex dispatch process      | Sensitivity analysis shows results robust to dispatch time                      |
| No hospital transport modeling               | Omits return-to-service dynamics                | Service time distribution absorbs this component                                |
| Independence assumption                      | Calls treated as independent                    | Reasonable for MVC; may understate correlation during major events              |

### 6.5 Future Research Directions

1. **Dynamic repositioning:** Extend P2 to time-varying allocations (different staging by shift)
2. **Road network integration:** Replace Haversine with OSRM/Google routing for more accurate travel times
3. **Multi-incident types:** Include medical emergencies, fires, and other call types
4. **Multi-borough optimization:** Scale from Manhattan to all 5 NYC boroughs

5. **Stochastic programming:** Account for demand uncertainty in the optimization model
  6. **Real-time decision support:** Develop dashboard for dispatchers with unit recommendations
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## 7. Conclusions and Recommendations

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### 7.1 Summary of Key Findings

1. **The current uniform allocation (P0) is dramatically suboptimal**, with only 64.4% 8-minute coverage and a mean response time of 8.08 minutes.
2. **The demand-weighted optimized allocation (P2) achieves near-perfect coverage (99.6%)** and reduces mean response time to 2.57 minutes—a 68.2% improvement.
3. **P2 is robust:** Performance rankings are invariant to demand fluctuations ( $\pm 100\%$ ), service time assumptions (20–30 min), and fleet size variations (15–40 units).
4. **The improvement is equivalent to tripling fleet capacity:** P2 with K=15 units outperforms P0 with K=40 units.
5. **All results are statistically significant** with large effect sizes (Cohen's  $d > .28$  for P0 vs P2), confirmed by ANOVA, Tukey HSD, and confidence interval analysis.

### 7.2 Implementation Recommendations

#### Immediate Actions (0–3 months)

- Adopt P2 allocation for K=20 units as the target staging plan
- Begin pilot deployment at 5 highest-impact firehouses (Midtown, Lower Manhattan)
- Establish KPI monitoring dashboard (mean RT, 8-min coverage, utilization)

#### Short-Term (3–6 months)

- Expand pilot to 15–20 firehouses based on initial results
- Calibrate travel time model with real dispatch data
- Integrate road network routing for improved accuracy

#### Medium-Term (6–12 months)

- Full deployment across all 48 Manhattan firehouses
- Develop time-of-day varying allocations (shift-specific P2)
- Extend model to other incident types

#### Long-Term (12+ months)

- Multi-borough optimization
- Real-time dynamic repositioning system
- Integration with CAD (Computer-Aided Dispatch) system

### 7.3 Expected Benefits

| Benefit                    | Estimate          | Basis                      |
|----------------------------|-------------------|----------------------------|
| Mean RT reduction          | −5.5 min per call | Simulation (n=30, p<0.001) |
| 8-min coverage improvement | +35.2 pp          | Simulation (n=30, p<0.001) |
| Effective fleet multiplier | 3×                | Fleet sensitivity analysis |
| Annual calls affected      | ~30,500 MVC calls | 3.48/hr × 8,760 hr         |
| Annual minutes saved       | ~167,750 min      | 30,500 × 5.5 min           |

## 8. References

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2. Daskin, M. S. (1983). A maximum expected covering location model: Formulation, properties and heuristic solution. *Transportation Science*, 17(1), 48–70.
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11. Pons, P. T., & Markovchick, V. J. (2002). Eight minutes or less: Does the ambulance response time guideline impact trauma patient outcome? *The Journal of Emergency Medicine*, 23(1), 43–48.
12. ReVelle, C., & Hogan, K. (1989). The maximum availability location problem. *Transportation Science*, 23(3), 192–200.



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## 9. Appendices

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### Appendix A: Mathematical Formulations

See `docs/optimization_formulation.md` for complete mathematical specifications of P0, P1, and P2 formulations, including decision variables, constraints, and solution properties.

### Appendix B: Simulation Model Specification

See `docs/conceptual_model.md` for the complete DES conceptual model, including entity definitions, event logic, and state transitions.

### Appendix C: Additional Statistical Tables

All statistical tables are available in `results/tables/`:

- `descriptive_statistics.csv` — Full descriptive statistics for all experiments
- `anova_results.csv` — Complete ANOVA results with assumptions tests
- `posthoc_comparisons.csv` — All pairwise comparisons with corrections
- `confidence_intervals.csv` — 95% CIs for all policy-metric combinations
- `effect_sizes.csv` — Cohen's d values for all comparisons

### Appendix D: Experimental Design Details

See `docs/experimental_design.md` for the complete factorial design specification, including factor levels, common random numbers (CRN) strategy, and warm-up period analysis.

### Appendix E: Verification & Validation Log

See `docs/verification_log.md` for detailed results of all 4 verification tests, 3 validation pilots, and 39 unit tests.

### Appendix F: Code Documentation

See `docs/code_documentation.md` for architecture overview, module descriptions, and extension guide. All source code is in `src/ems_readiness/` with 7,134 total lines across 14 modules.

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## 10. List of Figures

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The following figures are generated by the analysis pipeline and stored in `results/figures/`. Each figure is referenced in the relevant section of this report or in supporting documentation.

| #  | Filename                    | Caption / Description                                               |
|----|-----------------------------|---------------------------------------------------------------------|
| 1  | cbd_heatmap.png             | Heatmap of CBD-area crash demand density and fire-house locations   |
| 2  | cbd_response_comparison.png | Response time comparison between policies under CBD stress scenario |
| 3  | cbd_scenario_comparison.png | CBD vs. Manhattan-wide scenario performance comparison              |
| 4  | distance_matrix_heatmap.png | Heatmap of Haversine distance matrix (48 firehouses × 30 precincts) |
| 5  | exp1_policy_comparison.png  | Experiment 1: Policy comparison box plots (P0 vs. P1 vs. P2)        |
| 6  | exp2_fleet_sensitivity.png  | Experiment 2: Response time vs. fleet size by policy                |
| 7  | exp3_demand_sensitivity.png | Experiment 3: Response time vs. demand multiplier by policy         |
| 8  | exp4_service_robustness.png | Experiment 4: Response time vs. service time mean by policy         |
| 9  | fig_cbd_comparison.png      | CBD-focused robustness comparison across policies                   |
| 10 | fig_crash_heatmap.png       | Spatial heatmap of crash incidents across Manhattan precincts       |
| 11 | fig_daily_demand.png        | Daily crash demand patterns (day-of-week variation)                 |
| 12 | fig_demand_model_fit.png    | NHPP demand model fit diagnostics — observed vs. predicted rates    |
| 13 | fig_firehouses_map.png      |                                                                     |

| #  | Filename                      | Caption / Description                                                |
|----|-------------------------------|----------------------------------------------------------------------|
|    |                               | Map of 48 Manhattan FDNY firehouses used as candidate staging sites  |
| 14 | fig_hourly_demand.png         | Hourly crash demand distribution (24-hour profile)                   |
| 15 | fig_hourly_rates.png          | Calibrated NHPP hourly arrival rate factors                          |
| 16 | fig_policy_comparison.png     | Summary policy comparison across all key metrics                     |
| 17 | fig_precinct_demand.png       | Per-precinct crash demand distribution across Manhattan              |
| 18 | fig_precinct_density.png      | Precinct-level demand density choropleth map                         |
| 19 | fig_temporal_trends.png       | Long-term temporal trends in crash demand (2012–2026)                |
| 20 | fig_tradeoff_curve.png        | Response time vs. coverage trade-off curve across fleet sizes        |
| 21 | nhpp_arrivals_demo.png        | Demonstration of NHPP thinning algorithm arrival generation          |
| 22 | opt_allocation_comparison.png | Allocation comparison across optimization models (top 20 firehouses) |
| 23 | opt_inputs.png                | Optimization input visualization (travel times and demand weights)   |
| 24 | opt_sensitivity.png           | Optimization sensitivity analysis: objective value vs. fleet size    |
| 25 | project_summary_dashboard.png | Comprehensive project summary dashboard with key results             |
| 26 |                               |                                                                      |

| #  | Filename                         | Caption / Description                                                |
|----|----------------------------------|----------------------------------------------------------------------|
|    | pub_fig1_policy_comparison.png   | Publication-quality: Policy comparison (Figure 1)                    |
| 27 | pub_fig2_fleet_sensitivity.png   | Publication-quality: Fleet sensitivity analysis (Figure 2)           |
| 28 | pub_fig3_demand_robustness.png   | Publication-quality: Demand robustness analysis (Figure 3)           |
| 29 | pub_fig4_service_sensitivity.png | Publication-quality: Service time sensitivity (Figure 4)             |
| 30 | pub_fig5_performance_heatmap.png | Publication-quality: Performance heatmap across scenarios (Figure 5) |
| 31 | queue_comparison_by_policy.png   | Queue metrics comparison by policy (§5.8)                            |
| 32 | queue_heatmap.png                | Queue length heatmap across experiments and policies                 |
| 33 | queue_vs_demand.png              | Queue metrics vs. demand multiplier                                  |
| 34 | queue_vs_fleet_size.png          | Queue metrics vs. fleet size                                         |
| 35 | seasonal_decomposition.png       | Seasonal decomposition of monthly crash demand                       |
| 36 | seasonal_heatmap.png             | Monthly × day-of-week crash demand heatmap                           |
| 37 | seasonal_patterns.png            | Seasonal variation analysis (§5.9)                                   |
| 38 | service_time_distribution.png    | LogNormal service time distribution with empirical comparison        |
| 39 | tod_speed_factors.png            | Time-of-day speed factor profile (24-hour)                           |
| 40 | travel_time_by_tod.png           | Travel time distribution by time-of-day band                         |
| 41 | validation_p0_vs_p2.png          | Validation pilot: P0 vs. P2 directional comparison                   |

| #  | Filename                          | Caption / Description                                           |
|----|-----------------------------------|-----------------------------------------------------------------|
| 42 | validation_sensitivity_K.png      | Validation pilot: Response time sensitivity to fleet size K     |
| 43 | validation_sensitivity_demand.png | Validation pilot: Response time sensitivity to demand intensity |
| 44 | verification_toy_timeline.png     | Verification: Toy example event timeline trace                  |

#### Distance Comparison Figures (§5.10):

- 45. `results/distance_comparison/distance_matrices_heatmap.png` — Side-by-side Haversine vs Manhattan distance heatmaps
- 46. `results/distance_comparison/distance_scatter.png` — Scatter plot of Haversine vs Manhattan distances
- 47. `results/distance_comparison/distance_comparison_bar.png` — Performance comparison bar chart
- 48. `results/distance_comparison/distance_comparison_boxplot.png` — Replication distribution box plots

#### CBD-Focused Comparison Figures (§5.11):

- 49. `results/cbd_focused_comparison/cbd_focused_comparison.png` — CBD vs non-CBD response time and coverage comparison
- 50. `results/cbd_focused_comparison/allocation_comparison.png` — Unit allocation comparison between strategies
- 51. `results/cbd_focused_comparison/equity_tradeoff.png` — Equity-efficiency tradeoff scatter plot

#### Capacity Sensitivity Figures (§5.12):

- 52. `results/capacity_comparison/full_spectrum_summary.png` — Full-spectrum capacity sensitivity summary (cap 1-5)
- 53. `results/capacity_comparison/performance_vs_capacity_K20.png` — Performance vs capacity at K=20
- 54. `results/capacity_comparison/performance_vs_capacity_K40.png` — Performance vs capacity at K=40
- 55. `results/capacity_comparison/rt_heatmap_K20.png` — Response time heatmap by policy × capacity at K=20
- 56. `results/capacity_comparison/rt_heatmap_K40.png` — Response time heatmap by policy × capacity at K=40
- 57. `results/capacity_comparison/allocation_comparison_K20.png` — Allocation comparison at K=20
- 58. `results/capacity_comparison/allocation_comparison_K40.png` — Allocation comparison at K=40

#### Production V2 Figures (§5.13):

- 59. `results/production_v2/figures/mean_rt_vs_K.png` — Mean response time vs fleet size (V2)
- 60. `results/production_v2/figures/coverage_vs_K.png` — 8-minute coverage vs fleet size (V2)
- 61. `results/production_v2/figures/rt_distribution_K20.png` — Response time distributions at K=20 (V2)
- 62. `results/production_v2/figures/utilization_vs_K.png` — Utilization vs fleet size (V2)
- 63. `results/production_v2/figures/effect_sizes.png` — Effect sizes across fleet sizes (V2)
- 64. `results/production_v2/figures/allocation_map_K20.png` — Allocation map at K=20 (V2)

- 65. `results/production_v2/figures/allocation_map_K30.png` — Allocation map at K=30 (V2)
- 66. `results/production_v2/figures/allocation_map_K40.png` — Allocation map at K=40 (V2)

**Total: 66 figures** generated across EDA, optimization, simulation, alternative analyses, capacity sensitivity, Production V2, and publication workflows.

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## 11. List of Tables

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The following tables are generated by the analysis pipeline and stored in `results/tables/`. CSV files are used for data interchange; LaTeX ( `.tex` ) files are provided for publication-quality typesetting.

| #  | Filename                    | Caption / Description                                                       |
|----|-----------------------------|-----------------------------------------------------------------------------|
| 1  | anova_results.csv           | Full ANOVA results with F-statistics, p-values, and effect sizes            |
| 2  | cbd_comparison.csv          | CBD vs. Manhattan-wide performance comparison table                         |
| 3  | cbd_summary_all.csv         | Comprehensive CBD experiment summary across all scenarios                   |
| 4  | confidence_intervals.csv    | 95% confidence intervals for all policy-metric combinations                 |
| 5  | descriptive_statistics.csv  | Descriptive statistics (mean, std, min, max, quartiles) for all experiments |
| 6  | effect_sizes.csv            | Cohen's d effect sizes for all pairwise policy comparisons                  |
| 7  | exp1_summary.csv            | Experiment 1 summary: Policy comparison at K=20                             |
| 8  | exp2_pivot_rt.csv           | Experiment 2 pivot: Mean response time by policy × fleet size               |
| 9  | exp3_pivot_rt.csv           | Experiment 3 pivot: Mean response time by policy × demand multiplier        |
| 10 | exp4_pivot_rt.csv           | Experiment 4 pivot: Mean response time by policy × service time mean        |
| 11 | optimization_comparison.csv | Optimization model comparison (demand-weighted, p-median, maximal coverage) |
| 12 | posthoc_comparisons.csv     | Tukey HSD post-hoc pairwise comparisons with corrected p-values             |
| 13 | queue_anova.csv             | ANOVA results for queue metrics across experiments                          |

| #  | Filename                        | Caption / Description                                                   |
|----|---------------------------------|-------------------------------------------------------------------------|
| 14 | queue_statistics.csv            | Queue statistics (fraction queued, mean/max queue length) by experiment |
| 15 | seasonal_analysis.csv           | Monthly seasonal variation analysis with factors and statistical tests  |
| 16 | sensitivity_summary.csv         | Overall sensitivity analysis summary across all experiments             |
| 17 | table1_baseline_comparison.csv  | Publication Table 1: Baseline policy comparison                         |
| 18 | table1_baseline_comparison.tex  | Publication Table 1: LaTeX version                                      |
| 19 | table2_anova_summary.csv        | Publication Table 2: ANOVA summary                                      |
| 20 | table2_anova_summary.tex        | Publication Table 2: LaTeX version                                      |
| 21 | table3_pairwise_comparisons.csv | Publication Table 3: Pairwise comparisons                               |
| 22 | table3_pairwise_comparisons.tex | Publication Table 3: LaTeX version                                      |
| 23 | table4_sensitivity_summary.csv  | Publication Table 4: Sensitivity analysis                               |
| 24 | table4_sensitivity_summary.tex  | Publication Table 4: LaTeX version                                      |

#### Alternative Analysis Tables:



| #  | Filename                                              | Description                                  |
|----|-------------------------------------------------------|----------------------------------------------|
| 25 | results/distance_comparison/comparison_table.csv      | Haversine vs Manhattan simulation comparison |
| 26 | results/distance_comparison/allocation_comparison.csv | Allocation differences by distance metric    |
| 27 | results/cbd_focused_comparison/comparison_table.csv   | CBD-focused vs Manhattan-wide performance    |
| 28 | results/cbd_focused_comparison/allocations.csv        | CBD-focused vs Manhattan-wide allocations    |

#### Capacity Sensitivity Tables:

| #  | Filename                                               | Description                                                 |
|----|--------------------------------------------------------|-------------------------------------------------------------|
| 29 | results/capacity_comparison/full_comparison.csv        | Full capacity sensitivity comparison (all cap × K × policy) |
| 30 | results/capacity_comparison/simulation_results.csv     | Simulation results for capacity experiments                 |
| 31 | results/capacity_comparison/optimal_configurations.csv | Optimal capacity configurations by policy                   |
| 32 | results/capacity_comparison/allocation_statistics.csv  | Allocation statistics across capacity levels                |

#### Production V2 Tables:

| #  | Filename                                                    | Description                                      |
|----|-------------------------------------------------------------|--------------------------------------------------|
| 33 | results/production_v2/<br>tables/descriptive_statistics.csv | V2 descriptive statistics<br>(cap=2, P0-spatial) |
| 34 | results/production_v2/<br>tables/anova_results.csv          | V2 ANOVA results                                 |
| 35 | results/production_v2/<br>tables/pos-thoc_comparisons.csv   | V2 pairwise comparisons                          |
| 36 | results/production_v2/<br>tables/confidence_intervals.csv   | V2 95% confidence intervals                      |
| 37 | results/production_v2/<br>tables/effect_sizes.csv           | V2 Cohen's d effect sizes                        |
| 38 | results/production_v2/<br>tables/queue_statistics.csv       | V2 queue statistics                              |
| 39 | results/production_v2/com-<br>parison_with_v1.csv           | V1 vs V2 comparison table                        |

**Total: 39 table files** (20 CSV + 4 LaTeX + 4 supplementary CSV + 4 capacity CSV + 7 V2 CSV).

---

## 12. Reproducibility

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This section provides complete instructions for reproducing all results presented in this report.

## 12.1 Environment Specification

| Component        | Version / Value                        |
|------------------|----------------------------------------|
| Python           | 3.11.6                                 |
| Operating System | Ubuntu Linux                           |
| NumPy            | $\geq 1.24.0$                          |
| pandas           | $\geq 2.0.0$                           |
| SimPy            | $\geq 4.0.0$ (DES engine)              |
| PuLP             | $\geq 2.7.0$ (MIP solver, CBC backend) |
| SciPy            | $\geq 1.11.0$                          |
| matplotlib       | $\geq 3.7.0$                           |
| seaborn          | $\geq 0.12.0$                          |
| geopandas        | $\geq 0.14.0$                          |
| Shapely          | $\geq 2.0.0$                           |
| folium           | $\geq 0.15.0$                          |
| tqdm             | $\geq 4.65.0$                          |
| openpyxl         | $\geq 3.1.0$                           |

Full dependency list: `requirements.txt`

## 12.2 Random Seeds and Reproducibility Strategy

All stochastic components use **deterministic seeding** to ensure exact reproducibility:

| Component                     | Seed / Strategy                                                           | Configuration                                                                  |
|-------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------|
| <b>Base seed</b>              | 42                                                                        | configs/demand.yaml → simulation.seed ,<br>configs/simulation.yaml → seed_base |
| <b>Production experiments</b> | SEED_BASE = 42 ; replication i<br>uses seed 42 + i                        | scripts/<br>run_production_experiments.py (line 52)                            |
| <b>CBD experiments</b>        | SEED_BASE = 42 ; replication i<br>uses seed 42 + i                        | scripts/<br>run_cbd_experiment.py (line 53)                                    |
| <b>Verification tests</b>     | Fixed seed 42 for all 4 tests                                             | scripts/run_verification.py                                                    |
| <b>Validation pilots</b>      | Pilot 1-2: seed base 100 ; Pilot 3: seed base 200 ; Pilot 4:<br>300 + rep | scripts/<br>run_validation_pilots.py                                           |
| <b>PRNG algorithm</b>         | NumPy PCG64 Generator                                                     | Used throughout via<br>np.random.default_rng(seed)                             |

**Common Random Numbers (CRN):** Dedicated random number streams are used for (1) NHPP arrival generation, (2) precinct assignment (multinomial), and (3) service-time sampling. Stream offsets ensure that changing one component (e.g., service time distribution) does not alter the arrival sequence, enabling valid variance reduction across policy comparisons.

## 12.3 Configuration Files

| File                                   | Purpose                             | Key Parameters                                                                     |
|----------------------------------------|-------------------------------------|------------------------------------------------------------------------------------|
| <code>configs/demand.yaml</code>       | NHPP demand model parameters        | Base rate (3.48/hr), lambda table paths, replications (30), seed (42)              |
| <code>configs/service.yaml</code>      | Travel time and service time models | Speed (20 mph), TOD factors, LogNormal(25, 10), dispatch delay (1.5 min)           |
| <code>configs/optimization.yaml</code> | MIP optimization settings           | Unit counts [20,30,40,48], capacity (5), threshold (8 min), CBC solver             |
| <code>configs/simulation.yaml</code>   | DES engine configuration            | Horizon (168 hr), warmup (0), replications (30), seed base (42), threshold (8 min) |
| <code>configs/cbd_scenario.yaml</code> | CBD robustness experiment           | Demand surge (2×), increased service times, CBD boundary, seed base (42)           |

## 12.4 Step-by-Step Reproduction Instructions

```
# 1. Clone repository and set up environment
git clone <repository-url>
cd ems-optimization
python3 -m venv venv
source venv/bin/activate
pip install -r requirements.txt

# 2. Process raw data (requires raw data files in data/raw/)
python src/ems_readiness/data_processing.py
# Alternatively: make data

# 3. Run demand modeling
python scripts/demand_modeling.py

# 4. Run optimization comparison
python scripts/run_optimization_comparison.py

# 5. Run verification tests (4 tests)
python scripts/run_verification.py

# 6. Run validation pilots (3 pilots)
python scripts/run_validation_pilots.py

# 7. Execute production experiments (1,440 runs)
python scripts/run_production_experiments.py

# 8. Analyze production results
python scripts/analyze_production_results.py

# 9. Run CBD robustness experiment (330 additional runs)
python scripts/run_cbd_experiment.py

# 10. Run gap closure analyses
python scripts/analyze_queue_metrics.py
python scripts/analyze_seasonal_patterns.py

# 11. Generate publication figures
python scripts/generate_publication_figures.py
python scripts/generate_summary_dashboard.py

# 12. Run alternative analyses (distance metric comparison)
python scripts/generate_manhattan_distance_matrix.py
python scripts/run_distance_comparison_experiment.py --reps 10

# 13. Run alternative analyses (CBD-focused optimization)
python scripts/run_cbd_focused_optimization.py --reps 10
```

## 12.5 Data Requirements

| File                                             | Size   | Source                     | Notes                                 |
|--------------------------------------------------|--------|----------------------------|---------------------------------------|
| Mo-tor_Vehicle_Collisions_-_Crashes_20260223.csv | 536 MB | NYC Open Data              | 2.24M records; not in Git (too large) |
| FDNY_Firehouse_Listing_20260223.csv              | 15 KB  | NYC Open Data              | 219 firehouses; tracked in Git        |
| Po-lice_Precincts_20260223.csv                   | 3.6 MB | NYC Open Data              | 78 precincts; not in Git              |
| manhattan_boundary.geojson                       | 5 KB   | NYC Open Data              | Tracked in Git                        |
| cbd_boundary.geojson                             | 2 KB   | MTA Congestion Relief Zone | Tracked in Git                        |
| nyc_borough_boundaries.geojson                   | 650 KB | NYC Open Data              | Tracked in Git                        |

## 12.6 Expected Outputs

Successful reproduction generates:

- **44 figures** in `results/figures/`
- **24 table files** in `results/tables/`
- **Simulation logs** in `results/simulation/`
- All statistical results should match to within floating-point tolerance ( $< 10^{-6}$  relative error) when using identical seeds and Python/NumPy versions.